

**SUPSI**

# A Bayesian Approach to Temperature Analysis

From theory to practice - modeling real climate data

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Bayesian Data Analysis and Probabilistic Programming, D3A - SUPSI

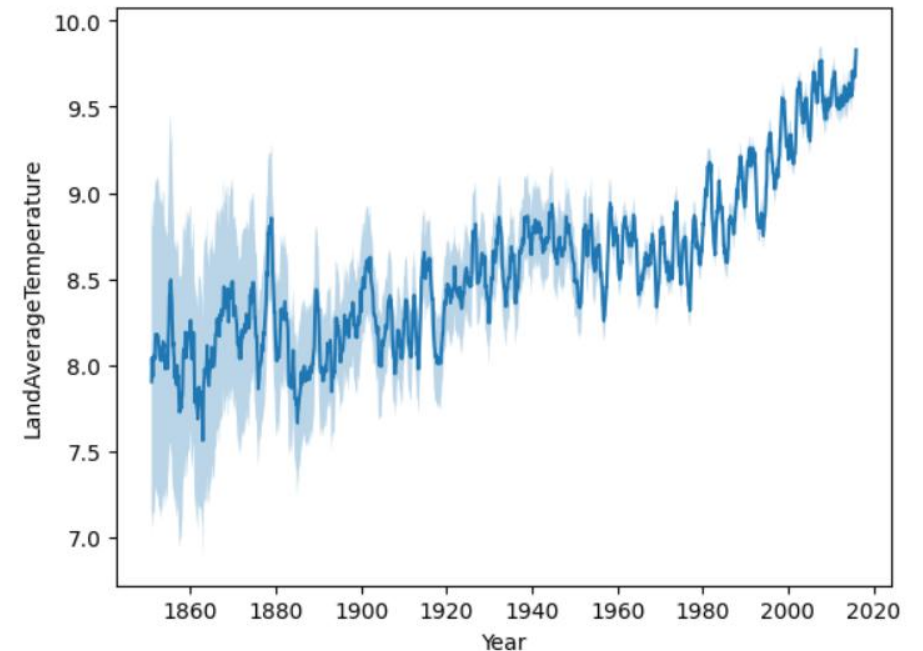
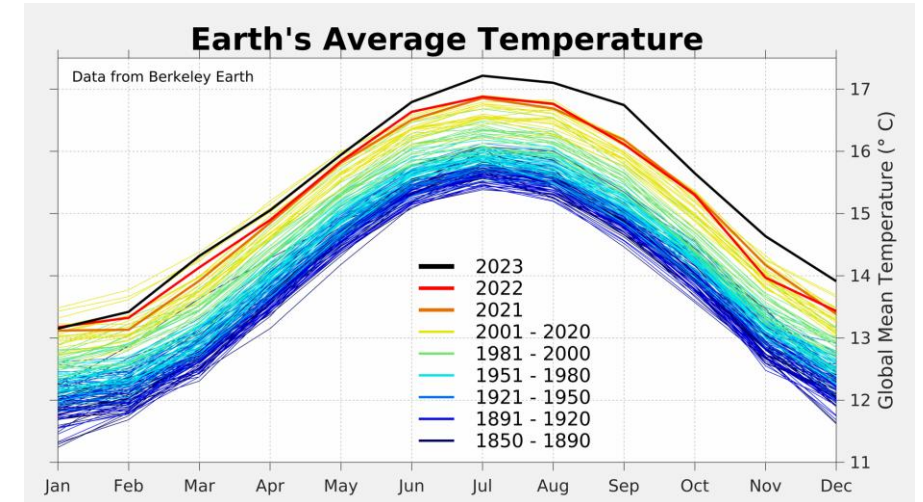
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# Dataset and motivations

## Dataset and Motivations

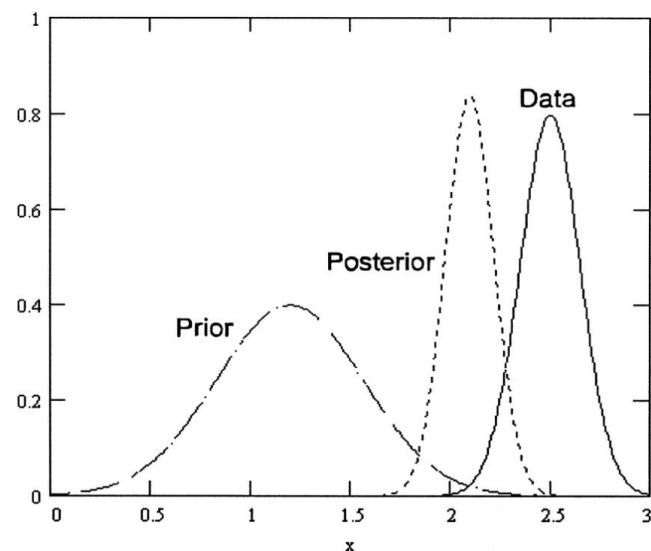
- Earth Surface Temperature Data from Berkley Earth
- Temperature measurements at city, state, country and global levels.
- Data subjected to measurement errors and time uncertainty
- Why using a bayesian approach?
  - Allows for the incorporation of **prior knowledge**, quite vast in this field
  - Provides **robust uncertainty quantification** to account for different types of noise
  - Let's us make **probabilistic statements** and better decisions
  - Can be extended to handle **hierarchical structures**



## Bayesian Modeling

- Model the parameters  $\theta$  that describe our data (average temperature, standard deviation, rate of change...)
- **Prior:** Assign a statistical distribution to each, encoding external knowledge
- **Likelihood:** Compare data generated by the model with the observed data  $Y$ .
- **Probabilistic Programming** (e.g. using PyMC) lets us write our mathematical models in code and obtain the posterior distributions for  $\theta$

$$p(\theta | Y) = \frac{\mathcal{L}(\theta) \cdot p(\theta)}{p(Y)}$$



# Hypothesis Testing

## First Hypothesis: Climatic differences between states

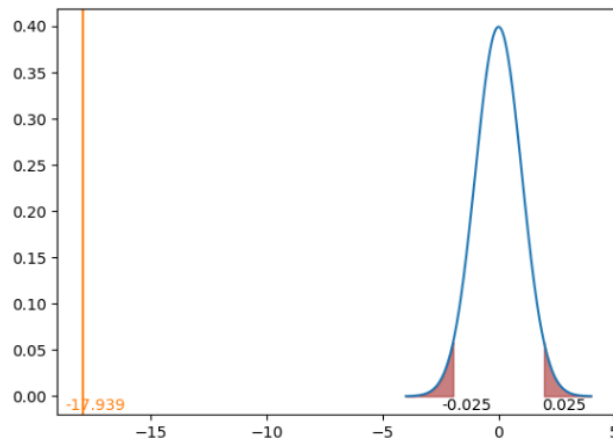
- California and Texas have very different climatic characteristics, despite being relatively close
- We want to determine whether the differences in average temperatures between these two states are statistically significant.

$$H_0 : \mu_{california} = \mu_{texas}$$

$$H_1 : \mu_{california} \neq \mu_{texas}$$

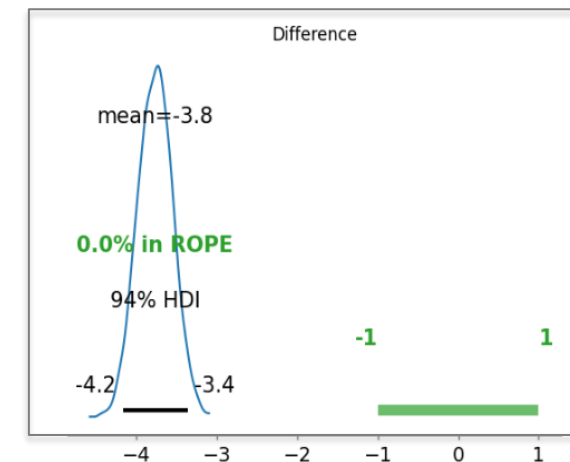
### Frequentist Approach

We perform a two-sample **t-test** to compare mean temperatures between the two states.



### Bayesian Approach

We use normal priors for means and half-normal priors for standard deviations, then check if the 94% posterior **HDI** contains zero or if the **ROPE** contains the HDI for practical equivalence.

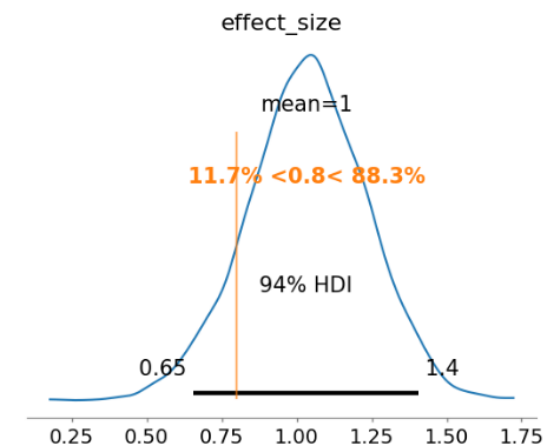
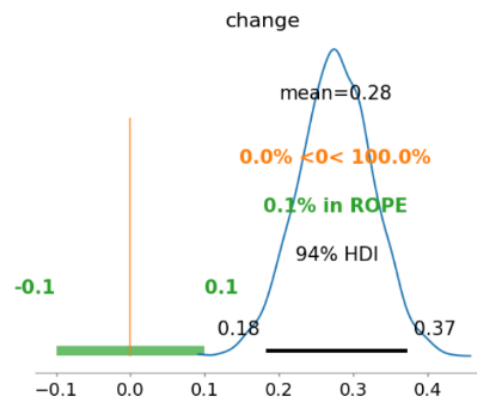
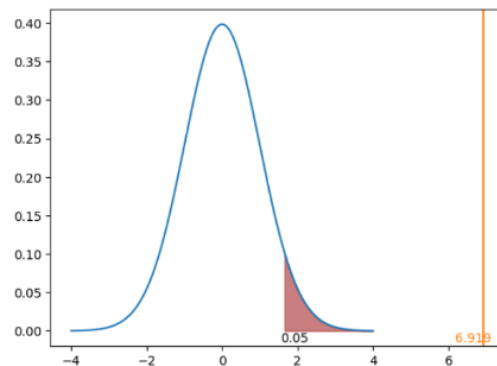


## Second Hypothesis: Mid-seasons over time

- We aim to test whether spring is warming faster than summer, potentially shrinking the seasonal temperature gap over time.
- Preliminary test:** Is there a significant difference between summer and spring temperatures? As expected, there is a significant difference.
- Main test:** Is this difference decreasing over time?
  - Compare temperature differences before and after 1900
  - Results:** Temperature gap is decreasing ( $t=6.9$ , posterior entirely  $> 0$ )
  - Effect Size:** 88% probability of large effect ( $>0.8$ ), confirming practical significance

$$H_0 : \mu_{pre\_1900} \leq \mu_{post\_1900}$$

$$H_1 : \mu_{pre\_1900} > \mu_{post\_1900}$$





# Bayesian Regression

## Latitude-based Regression

- Predict the temperature of cities, using only latitude as a feature
- The linear regression formula for a model with a single feature  $x_1$  is  $Y = \alpha + \beta x_1 + \epsilon$ , where  $\epsilon \sim N(0, \sigma^2)$  is measurement noise.
- Each of the parameters  $\alpha, \beta, \sigma$  can be modeled.
- **Prior knowledge:**
  - Average annual temperatures in equatorial lands ( $x_1 = 0$ ) are around 27C, then decrease by 0.6 C per latitude degree
- Results:  $R^2 = 0.24$  (latitude alone insufficient)

**Priors:**

$$\alpha \sim N(27, 3^2)$$

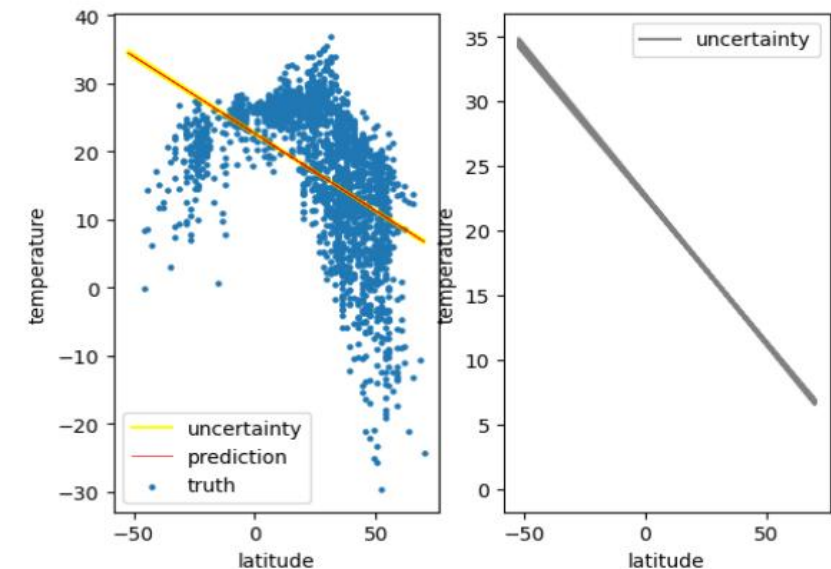
$$\beta \sim N(-0.6, 0.1^2)$$

$$\sigma \sim \text{HalfNormal}(7.5)$$

**Likelihood:**

$$Y \sim N(\alpha + \beta x_1, \sigma^2)$$

Predictions with Epistemic Uncertainty



## USA city temperature

- Features: latitude, longitude, year and month.
- Since prior knowledge is often more accessible when predictors are at their average values, we center the predictors so that the intercept changes from  $\alpha = \mathbb{E}[y \mid x = 0]$  to  $\alpha_c = \mathbb{E}[y \mid x = \bar{x}]$
- **Prior knowledge:**
  - **Year:** Weak prior assuming it has no effect on temperature.
  - **Latitude & Longitude:** Prior reflects south–north temperature decrease, while east–west variation is irregular and uncertain.
  - **Month:** two possible approaches, one-hot (shown) and sine–cosine representations with neutral priors.
- $R^2 = 0.89$

### Priors:

$$\alpha_c \sim N(11.1, 1.4)$$

$$\beta_{\text{year}} \sim N(0, 1)$$

$$\beta_{\text{lat}} \sim N(-0.56, 0.15)$$

$$\beta_{\text{lon}} \sim N(0, 0.1)$$

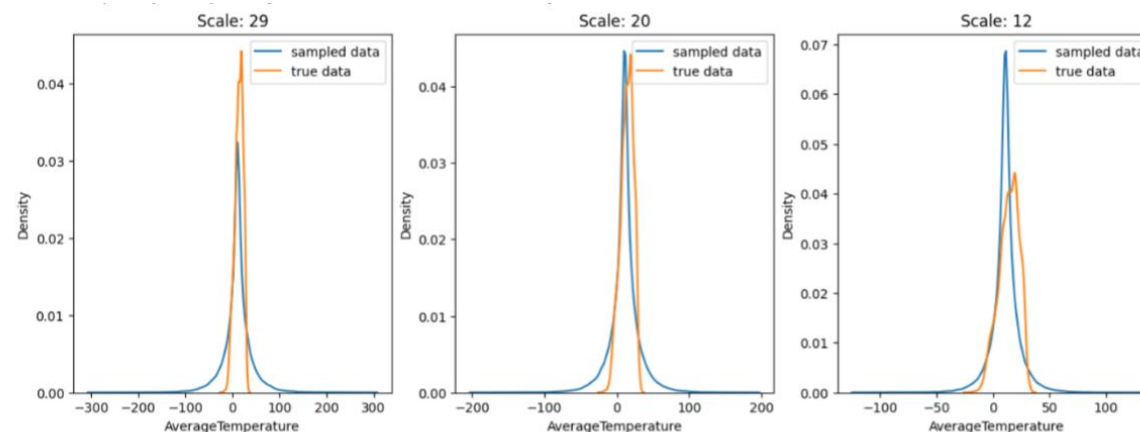
$$\beta_{\text{month}, m} \sim N(0, 1), \quad m = 1, \dots, 11$$

$$\sigma \sim \text{HalfNormal}(12)$$

### Likelihood:

$$\mu_i = \alpha_c + \beta_{\text{year}} \cdot \text{year}_i + \beta_{\text{lat}} \cdot \text{lat}_i + \beta_{\text{lon}} \cdot \text{lon}_i + \sum_{m=1}^{11} \beta_{\text{month}, m} \mathbf{1}(\text{month}_i = m)$$

$$y_i \sim N(\mu_i, \sigma^2)$$



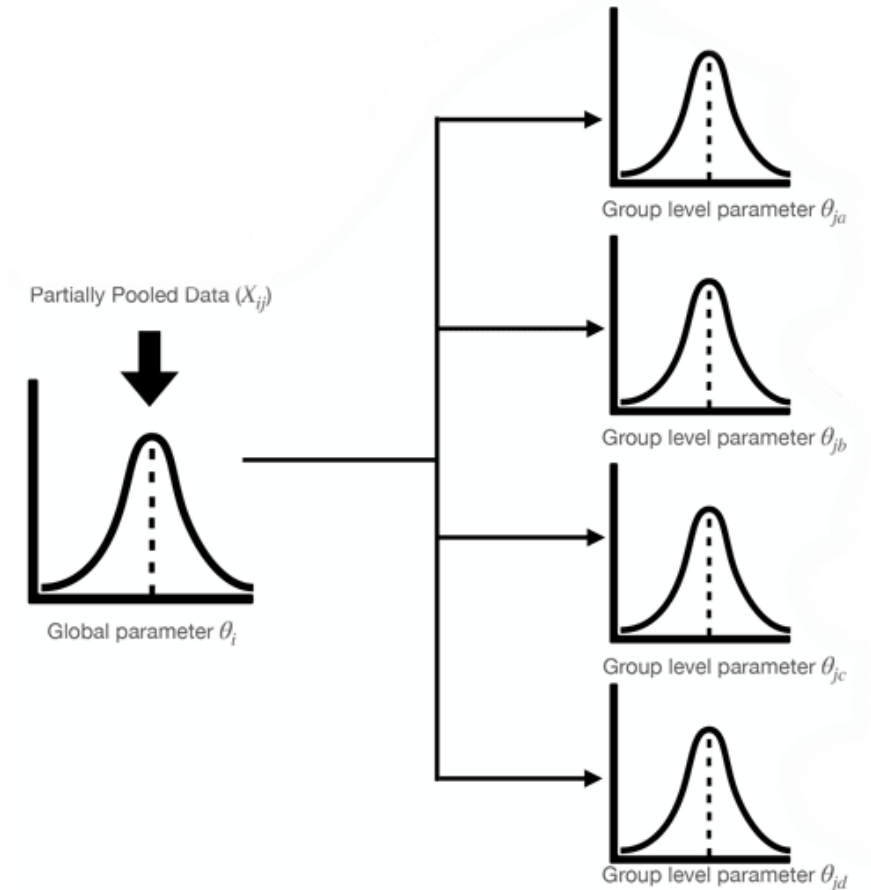
# Hierarchical Modeling

## Hierarchical Modeling

We may have multiple datasets representing different but related groups (e.g. cities).

Three modeling approaches:

- **Pooled:** All cities have same temperature distribution
- **Unpooled:** Each city is completely independent
- **Partially Pooled (Hierarchical):** Cities are related by drawing their individual parameters from a shared population distribution.
  - Borrows strength across groups
  - Shrinkage: extreme values pulled toward population mean
  - Can predict for new, unseen cities



## Pooled and Unpooled models

- **Pooled model:** temperature mean is defined like in the previous model.
  - **Uncertainty in the estimate** of  $\mu$  is 1.4, reflecting how confident we are about the overall mean.
  - $\sigma$  captures the **spread of temperatures across all cities**, assuming a single shared distribution.
- **Unpooled model:** each city has its own mean temperature, denoted by  $\mu_j$ .
  - We're estimating many separate means, thus, the **uncertainty in each**  $\mu_j$  is larger (2.8) than in the pooled case (1.4).
  - $\sigma_{temp}$  reflects the **spread of temperatures within each individual city**, not across cities.

### Priors:

$$\mu \sim N(11.1, 1.4^2)$$

$$\sigma \sim \text{HalfNormal}(12)$$

### Likelihood:

$$y_{ij} \sim N(\mu, \sigma^2)$$

### Priors:

$$\mu_j \sim N(11.1, 2.8^2)$$

$$\sigma_{temp} \sim \text{HalfNormal}(8)$$

### Likelihood:

$$y_{ij} \sim N(\mu_j, \sigma_{temp}^2)$$

## Hierarchical Model

The model can be seen as a mid-point between the pooled and unpooled models.

- We assume a **shared population distribution** with mean  $\mu_{pop}$  and spread  $\sigma_{pop}$ .
- Each city  $j$  has a **different mean**  $\mu_j$  drawn from the population distribution.
- The **within-city temperature spread**  $\sigma_{temp}$  is typically smaller than  $\sigma_{pop}$ .

Since we estimate the population-level parameters, we can then sample from the distribution to obtain a **new group**  $\mu_j$  and make predictions for that group.

### Hyperpriors:

$$\mu_{pop} \sim N(11.1, 1.4^2)$$

$$\sigma_{pop} \sim \text{HalfNormal}(12)$$

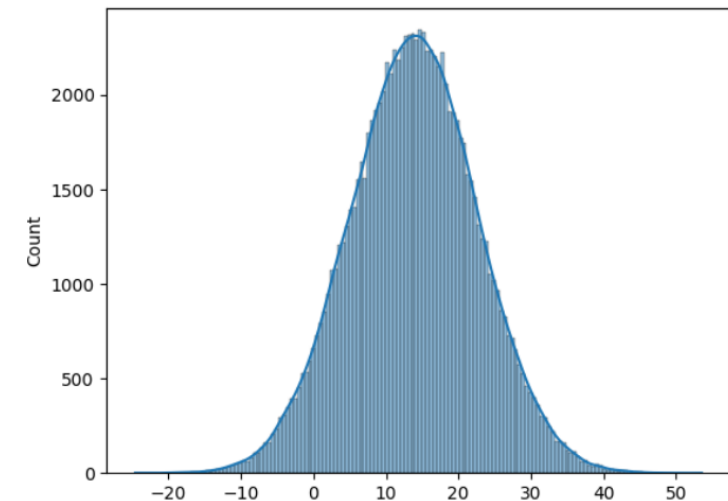
### Priors:

$$\mu_j \sim N(\mu_{pop}, \sigma_{pop}^2)$$

$$\sigma_{temp} \sim \text{HalfNormal}(8)$$

### Likelihood:

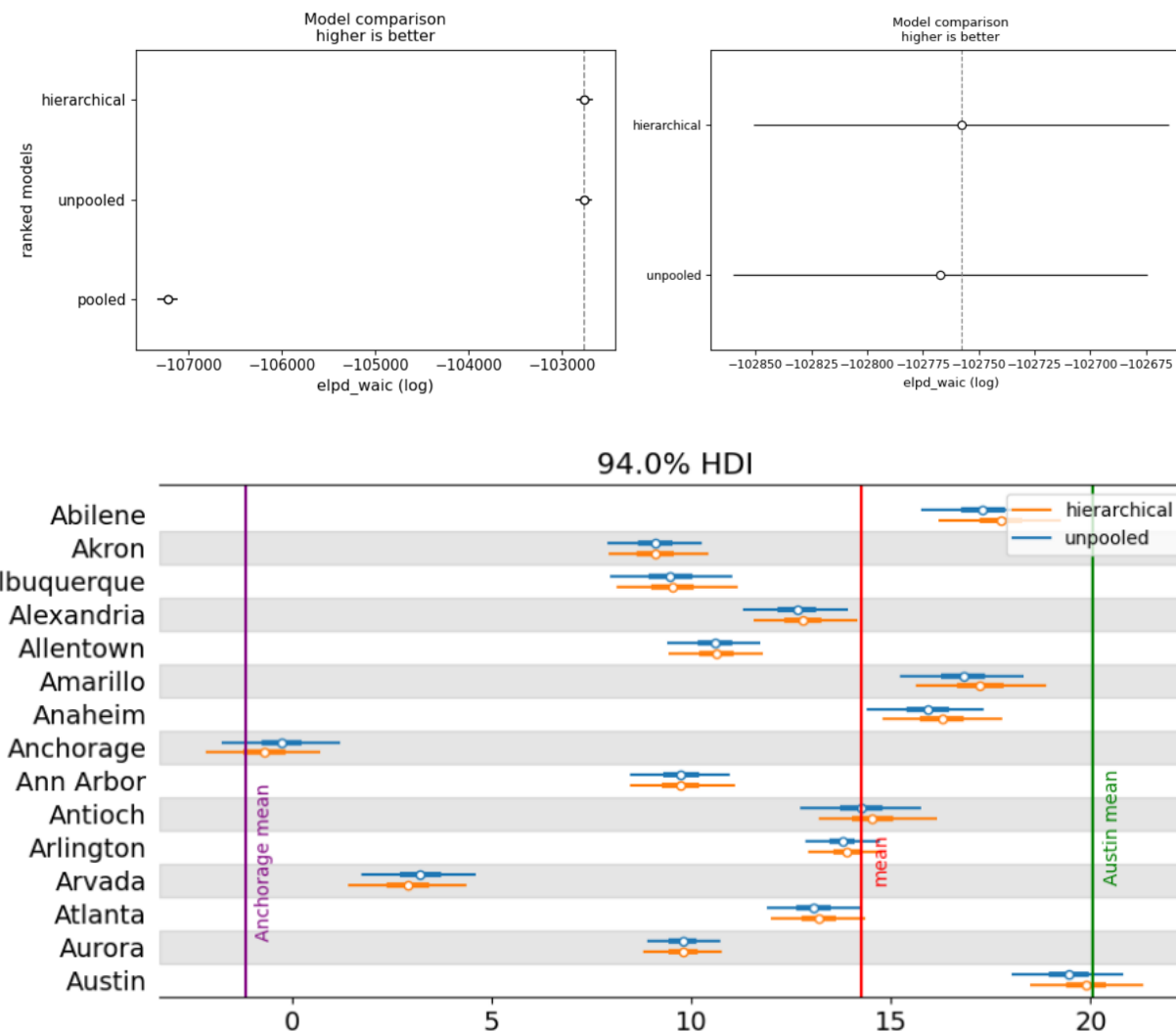
$$y_{ij} \sim N(\mu_j, \sigma_{temp}^2)$$



Predictive distribution for a new city

## Comparing levels of pooling

- The **pooled model** performs poorly, it completely ignores variability across groups, treating all data as if it came from a single source.
- The **hierarchical model** yields better predictive performance than the **unpooled model**, as measured by **WAIC**, an indicator of how well the model generalizes to unseen data.
- While **unpooled estimates** are more sensitive to noise due to limited data within each group, **hierarchical estimates** benefit from partial pooling, being more informed and regularized.





Thank you

Questions?